

Parameterized Model Order Reduction for Fast Transient Electromagnetic Simulations

Mohammad Abid Bazaz, Mashuq-un-Nabi and S. Janardhanan

Abstract— Transient Electromagnetic problems constitute an area of significant investigative effort. The principal computational issue in these problems is the solution of large system of differential algebraic equations (DAE's) obtained after Finite Element (FE) discretization. Model Order Reduction (MOR) Techniques provide a mechanism to generate reduced order models from the detailed description of the original FE network. This is achieved by using moment matching techniques, where the reduced order model matches the moments of the original system to approximate the response with a low order transfer function. However, these numerical techniques all conserve the original system moments only with respect to frequency. While this provides a significant CPU cost advantage when performing a single frequency sweep, a new reduced order model is required each time a parameter is varied in the structure under study. This necessitates the use of parametric MOR strategies so as to expedite optimization and design space exploration cycles. In this work, we present a methodology for transient electromagnetic field simulations through parameterized model order reduction (pMOR). The proposed methodology is illustrated for a generic system with promising results and a significant saving in computational effort.

Index Terms-- Differential Algebraic Equations; Finite Element Modeling; moments; Model Order Reduction; parameters

I. INTRODUCTION

TRANSIENT electromagnetic problems constitute an area of significant investigative effort [1][2][3]. They occur in a wide range of applications, from simulation of transient electrostatic phenomena and transient eddy current systems [2], to simulation of electrical machines under transient or field conditions [3][4]. These electromagnetic phenomena are governed by partial differential equations (PDE's). Analytical methods for solving PDE's are available only for simple geometries [5]. As such, numerical methods like Finite Element Method (FEM), Finite Difference Method (FDM) and Boundary Element Method (BEM) become inevitable for the modeling of electromagnetic fields over complicated

domains [6][7][8]. FEM is generally preferred for its reliability and applicability to a broad class of engineering areas [6][9]. For transient field computations, the FE models are large systems of DAE's generally involving sparse, symmetric and often positive-definite coefficient matrices. These DAE's are normally solved through various time stepping schemes wherein each step requires solution of a large system of algebraic equations of same size as the number of DAE's in the FE model.

In [10], a two-tier strategy for Krylov-based reduction of field-circuit coupled systems was proposed by the authors. Finite Element method was used for modeling the electromagnetic field. As a first stage of reduction, the method proposed in [9] was used for converting the large system of DAE's governing the field, obtained after FE discretization, into a pure Ordinary Differential Equation (ODE) model, by exploiting the structure of the coefficient matrices of the FE model, with the parameters symbolically preserved in the ODE model. As a second stage reduction, Krylov-based MOR strategy was used for reducing the pure ODE model at specific instances of the parameter values. However, although the ODE model obtained after first stage reduction was a parameterized model, the reduced order model after the second stage reduction was a non-parameterized one. As such, for every new value of the parameter, we need to carry out the reduction of the ODE model corresponding to that specific value of the parameter, which is computationally cumbersome leading to strenuous design and optimization cycles. In this paper, we have extended our previous work by extracting a parameterized reduced order model from the parameterized ODE model, obtained after first stage reduction, by common projection matrix approach [11]. With this modification, we are able to preserve the parameters symbolically in the reduced order model. Therefore, any new computation of the field requires a solution of equations of significantly smaller size without repeating the reduction step. Numerical simulation results of the proposed scheme are shown for a generic electromagnetic system.

The paper is organized as follows. The following section recalls the approach used in [10] for non-parameterized reduction of the field. In section III, the approach suggested for parameterized model order reduction is presented. Numerical implementation of the proposed modeling strategy is illustrated for a typical electromagnetic system in Section IV, followed by a conclusion in the last section.

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II. NON-PARAMETERIZED REDUCTION OF THE FIELD

A. Finite Element Modeling of the field

The problem domain Ω , enclosed by the overall Dirichlet boundary Γ_d , includes air region without eddy currents (Ω_1) and eddy current region (Ω_2) as shown in Fig.1. The interface between Ω_1 and Ω_2 is denoted by Γ_{12} . The eddy current domain may consist of k solid current carrying regions. The time-varying eddy current equation over Ω_2 is given by:

$$\nabla \times \left(\frac{1}{\mu} \nabla \times A \right) = -\sigma \left(\frac{\partial A}{\partial t} + g \right) \quad (1)$$

where A is the magnetic vector potential, $\nu = 1/\mu$ is the reluctivity, σ the electrical conductivity, g is the potential gradient normal to the plane applied external to the conducting region.

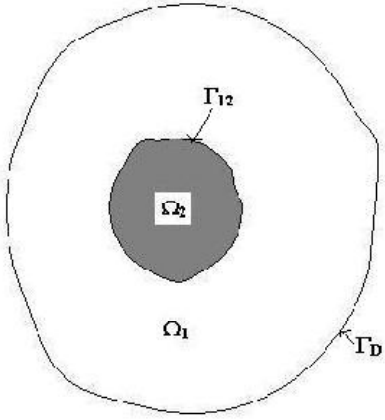


Fig 1: The problem domain

Following standard FE procedure, the global FE model of the field is obtained as

$$\left[K + S \frac{d}{dt} \right] \bar{A} = \bar{f} \bar{g} \quad (2)$$

where $\bar{A}$ is a vector of nodal magnetic potentials and $\bar{g}$ is a vector of nodal electric potentials applied over the conducting region, while K , S and $\bar{f}$ are FE related coefficient matrices.

The structure of the coefficient matrices in the left hand side of Eq. (2) can be identified as

$$K = \begin{bmatrix} D_1 & D_2 & 0 \\ D_2^T & D_3 + F_1(\nu) & F_2(\nu) \\ 0 & F_2^T(\nu) & F_3(\nu) \end{bmatrix} \quad (3)$$

$$S = \begin{bmatrix} 0 & 0 & 0 \\ 0 & E_1(\sigma) & E_2(\sigma) \\ 0 & E_2^T(\sigma) & E_3(\sigma) \end{bmatrix} \quad (4)$$

The three blocks of rows and columns correspond to nodes inside Ω_1 (air), on interface Γ_{12} and inside Ω_2 respectively. Following the method illustrated in [7], the time varying finite element model of the electromagnetic field can be obtained as

$$[M_{21} + \sigma M_{22} \frac{d}{dt} + \nu M_{23}] \bar{y}(t) = -\sigma f_2 \bar{g}(t) \quad (5)$$

where the matrices M_{21} , M_{22} and M_{23} were derived from K and S , while f_2 contains the rows of $\bar{f}$ corresponding to the rows

in Ω_2 . The whole vector $\bar{A}(t)$ of nodal potentials can be obtained from $\bar{y}(t)$ through the equation $\bar{A}(t) = P \bar{y}(t)$ where the columns of matrix P form a basis for the kernel of $[D_1 \ D_2 \ 0]$. It may be noted that the model (5) is a pure ODE model whose size is same as the number of nodes in the conducting region, which is far less than the original size of the FE model. Furthermore, as mentioned in the introductory section, the parameters are preserved symbolically, and thus the model as given by eq. (5) qualifies as a parameterized model. Thus, having calculated P once, any new computation of the field requires the solution of a much smaller system of ODE's.

B. Krylov based reduction of ODE model

Even after first stage reduction, the size of the ODE model to be solved is still significantly large, as such Krylov based reduction was proposed as a second stage reduction. Although the model obtained after second stage reduction faithfully captures the dynamic characteristics of model described by eq. (5), but the parameterized nature of the model gets destroyed. The model obtained after second stage reduction has the form:

$$E_r \dot{z} = A_r z + B_r u \quad (6)$$

where $E_r = V^T E V$, $A_r = V^T A V$ and $B_r = V^T B$. The matrices E , A , B are obtained from (5) for some specific values of σ and ν , and V is the orthogonal basis of the Krylov subspace obtained through Arnoldi algorithm. For detailed discussion on Krylov based reduction, the readers may refer to [12].

III. PARAMETERIZED MODEL ORDER REDUCTION

Typically, a reduced order model represents a specific instance of the physical system under study. Parametric variations in the physical model necessitate the generation of new reduced models. The benefits of model reduction will be fully realized only if the parametric dependence found in the original dynamical system can be passed on to the reduced model. This precisely is the goal of Parametric Model Order Reduction (pMOR); to find a reduced model that preserves the parameter dependency allowing a variation of any of the parameters without the need to repeat the reduction step. Multi-variate Pade approximation based methods have been used for parametric model order reduction both with implicit and explicit moment matching [13][14]. Other approaches that have been used for pMOR include interpolation of the transfer function [15] and reduced basis methods [16]. However, in this work, we have used the common projection matrix approach proposed in [11].

In the context of transient electromagnetic simulations, we need to reduce the model governed by Eq. (5) while preserving the parameters σ and ν symbolically in the reduced model. The strategy for obtaining the parameterized reduced order model is discussed in the next subsection.

A. pMOR strategy used: Common Projection Approach

Consider the system formulation as shown in Eq. (5). If the value of σ and ν is fixed, the equation can be written as:

$$E \dot{x} = A x + B u \quad (7)$$

where $E = \sigma M_{22}$, $A = -(M_{21} + \nu M_{23})$, $B = -\sigma f_2$ and $u = \bar{g}(t)$

Using the Arnoldi process, it is possible to find a Krylov subspace that could be used to reduce the system for these fixed values of the system parameters. However, this process by itself does not generate a parametric model. In order to obtain a model that is valid over a range of values for the input parameters, first compute several of these Krylov subspaces $V_1, V_2, \dots, V_m$ corresponding to different values of σ and v . These m subspaces contain valuable information about the system for different values of the parameters. This information can be merged by forming the matrix

$$V_{all} = [V_1 \ V_2 \ V_3 \ \dots \ V_m] \quad (8)$$

Since there are generally similarities between the individual projection matrices, the matrix V_{all} contains redundant information. For efficiency, the redundant information is filtered and the matrix is ortho-normalized using Singular Value Decomposition (SVD). It computes the matrices U , S and V from V_{all} such that

$$V_{all} = USV^T \quad (9)$$

where S is a diagonal matrix with non-negative elements and in decreasing order and U , V are unitary matrices. The values on the diagonal of S provide a measure of the relative importance of the various vectors in the ortho-normal subspace defined by the columns of U . This provides a convenient way to truncate the redundant subspace and trade-off accuracy versus size of the reduced order model. Thus, taking only the columns of U that correspond to large values in S will give us a compact subspace V_{com} . This approach retains all the essential information contained in the original subspaces and provides an efficient parametric reduced order model. The parametric reduced order model obtained has the form:

$$(M_{21r} + \sigma M_{22r} d/dt + v M_{23r})z = -\sigma f_{2r}g \quad (10)$$

where $M_{21r} = V_{com}^T M_{21} V_{com}$, $M_{22r} = V_{com}^T M_{22} V_{com}$, $M_{23r} = V_{com}^T M_{23} V_{com}$, $f_{2r} = V_{com}^T f_2$.

This approach provides a convenient way of trade off between speed and accuracy as the vectors in U are ordered by their relative contributions to the essential subspace. While this approach requires more computational cost than traditional Krylov methods for the construction of the subspace, it has the advantage of reducing the size of the model which is beneficial when this model is used repetitively in the simulation such as in case of design optimization.

IV. NUMERICAL RESULTS

The proposed framework is tested on a plate-bar geometry shown in Fig. 2 wherein repetitive field computations are to be performed for different values of the parameters σ and v . The bars and the eddy current region together constitute Ω_2 , while the remaining air region constitutes Ω_1 , surrounded by the boundary Γ_d . The FE code was developed using Scilab computational package which gave a DAE system of size 5665 excluding the boundary nodes. Out of these 5665 equations, 4628 equations were identified as the algebraic equations and the remaining 1037 as the differential equations. Using the first stage of reduction, DAE system of size 4628 was converted into a pure ODE system of size 1037. It is to be mentioned here that the pure ODE model obtained after first stage reduction is a parametric model with the parameters σ

and v symbolically preserved. For some specific values of σ and v , the ODE model obtained after first stage reduction was

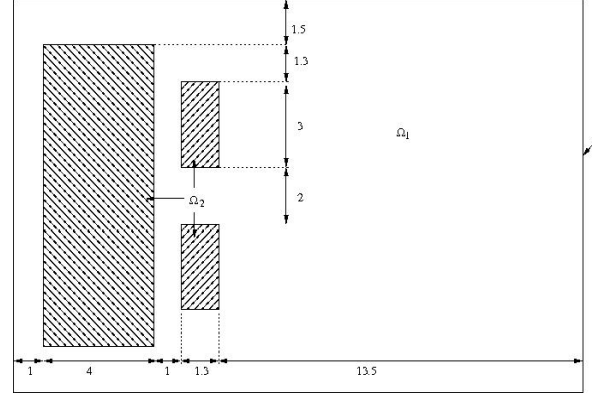


Fig (2): Plate-bar Geometry

directly reduced via conventional Krylov-subspace based method and the non-parametric reduced order model of size 12 was found to well approximate the original ODE model.

Using the approach presented in section III, projection matrix V_{com} was obtained by the aggregation of several projection matrices obtained at discrete values of the parameters. The orthogonal basis for the same was obtained after the SVD step and after fixing a certain tolerance for the singular values, the size of V_{com} was found to be 1037x19.

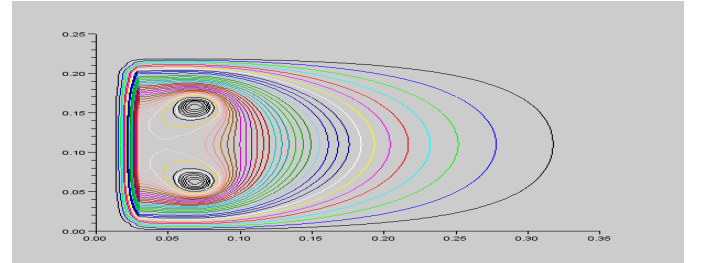


Fig (3): Snapshot of the field obtained from DAE model (Size: 5665)

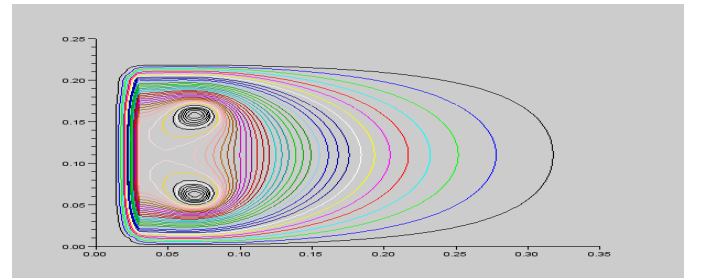


Fig (4): Snapshot of the field obtained from non-parametric reduced model (Size: 12)

The congruence transformation using this projection matrix gave a parametric reduced order system of size 19. The relative sizes of the systems to be solved for electromagnetic field computations is an index of saving in computational resources achieved through the proposed framework.

The original large sized DAE model was time-stepped through backward difference time stepping and the snapshots of the field profile were obtained (Fig: 3). The snapshots from the directly reduced non-parametric model (Fig: 4) and the

parametric model (Fig: 5) were also obtained and were found to exactly match the field profile snapshots obtained from the

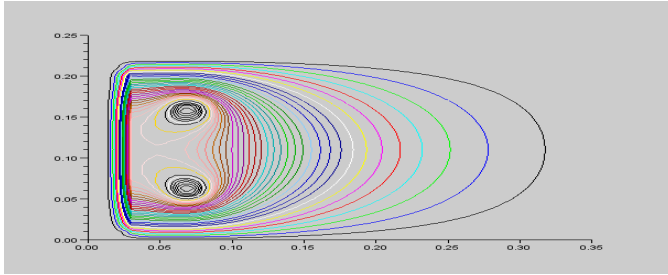


Fig (5): Snapshot of the field obtained from parametric reduced model (Size: 19)

original DAE model. Although the size of the directly reduced non-parametric model (size: 12) is less than that of the parametrically reduced model (size: 19), but the advantages of the parametric reduction far overweigh the slight increase in the size of the model. The parameters are symbolically preserved in the parametric model and as such can be used for simulating for different values of parameters without repeating the reduction step. The evolution of the field for a representative node also shows excellent matching between the directly reduced non-parametric model and the parametrically reduced model as is evident in Fig (6).

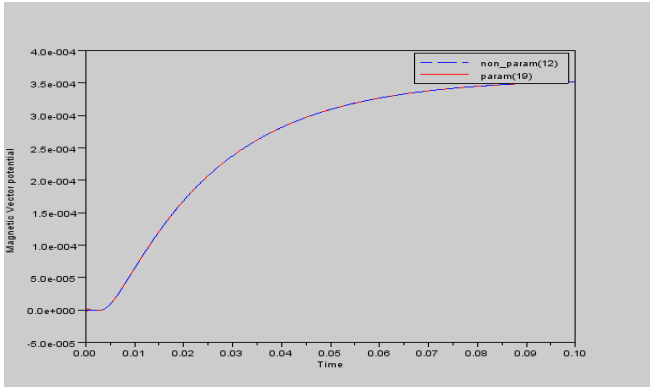


Fig (6): Time History of a representative node

V. CONCLUSION

A framework for parametric model order reduction for transient electromagnetic computations has been presented and numerically validated. The main advantage is that any new simulation of the field, for a different set of parameters, does not require the reduction step to be repeated. The parameters are symbolically preserved in the reduced model and as such any new simulation requires the time-stepping of same set of reduced number of equations for the required parameter values. The viability of coupling the parametric reduced order model with a circuit for coupled field-circuit formulations is being explored as a future direction of work.

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